

Production Deployment Fix Guide

Issue Summary

The backend server is failing to start because:

1. PM2 configuration had incorrect directory paths (`/var/www/gema` vs `/var/www/GEMA-Project`)
2. The TypeScript code needs to be compiled to JavaScript before running
3. PM2 is trying to run TypeScript files directly instead of compiled JavaScript files

What Was Fixed

- Updated `ecosystem.config.js` to use correct paths: `/var/www/GEMA-Project/backend`
- Both main app and worker paths have been corrected

Steps to Fix Production Server

1. SSH into your production server

```
ssh root@your-server-ip
```

2. Navigate to the project directory

```
cd /var/www/GEMA-Project
```

3. Pull the latest code

```
git fetch origin
git checkout backend_auth # or your main branch
git pull origin backend_auth
```

4. Navigate to backend directory

```
cd backend
```

5. Install dependencies (if needed)

```
npm install
```

6. Build the TypeScript code

```
npm run build
```

This compiles TypeScript files from `src/` to JavaScript in `dist/`

7. Stop and delete the current PM2 process

```
pm2 stop kidrove
pm2 delete kidrove
```

8. Start the server with the correct ecosystem config

```
pm2 start ecosystem.config.js --env production
```

9. Save the PM2 configuration

```
pm2 save
```

10. Check the server status

```
pm2 status
pm2 logs gemma-backend --lines 50
```

Expected Result

You should see:

- Two processes running: `gema-backend` and `gema-worker`
- Both should be in "online" status
- No module resolution errors in the logs

Alternative: Quick Restart (if paths are already correct)

If you just need to rebuild and restart:

```
cd /var/www/GEMA-Project/backend
npm run build
pm2 reload ecosystem.config.js --env production
```

Verification

Test the API is working:

```
curl http://localhost:5001/health
```

Should return a healthy status response.

Common Issues

If build fails

- Check Node.js version: `node --version` (should be `>= 20.0.0`)
- Check npm version: `npm --version` (should be `>= 10.0.0`)
- Clear node_modules and reinstall: `rm -rf node_modules && npm install`

If PM2 won't start

- Check logs: `pm2 logs`
- Verify `.env` file exists in backend directory
- Check port 5001 isn't already in use: `lsof -i :5001`

If module errors persist

- Ensure you're in the correct directory: `/var/www/GEMA-Project/backend`
- Verify dist folder exists and has content: `ls -la dist/`
- Check `dist/config/index.js` exists: `ls -la dist/config/`